

DSM050 Data Visualisation Mid Term CourseWork:

What factors explain sports success in FIFA World Cups?

Introduction

The FIFA World Cup is widely recognized as the world's largest sports event, bringing together national teams, fans, and audiences from virtually every region of the world. Beyond its sporting significance, the tournament represents a global cultural phenomenon that attracts unprecedented levels of public engagement. According to FIFA, the 2022 FIFA World Cup generated engagement from more than five billion people across television, digital platforms, and social media (FIFA, 2023). Given its worldwide popularity and the wealth of historical data available, the FIFA World Cup provides an ideal context for exploring the factors associated with team performance through data visualization and analytical techniques.

There is a wide variety of data available for events such as the FIFA World Cup, ranging from audience statistics and player performance to financial investments, economic returns, match results, champions, and much more. This project focuses on a set of research questions that provide interesting insights from a broader perspective. The objective is to present these questions as compelling stories that could be featured in a magazine or digital publication, using data visualizations that effectively communicate the findings while being supported by appropriate context and interpretation.

Research Context and Problem Definition

When attempting to explain why certain national teams consistently outperform others, a wide range of hypotheses emerges. Some of the most common explanations include economic development, cultural and historical factors, the popularity of football within a country, investment in player development and infrastructure, and the quality of domestic leagues. In reality, international football success is likely the result of a combination of these factors, together with others that are difficult to observe or quantify. Given the scope of this study, the analysis is limited to performances in the FIFA World Cup rather than international football as a whole. Batarfi and Reade (2021) found that GDP per capita has a significant positive effect on the winning proportion of national football teams in international competitions. Building on their findings, this study examines whether a similar relationship exists specifically in FIFA World Cup tournaments.

Aside from economic factors, home advantage is another widely discussed explanation for differences in sporting performance. Nevill and Holder (1999) identify four main categories of factors that contribute to home advantage: crowd support, familiarity with the venue, travel, and rule factors. Although the relative importance of these factors varies across sports, each

has been shown to influence athletic performance. Familiarity refers to the advantage of competing in well known facilities and environmental conditions. Travel encompasses the physical and psychological demands associated with long distance travel, including fatigue and time-zone changes. Rule factors include sport specific regulations or competition structures that may favor the home team in certain disciplines. Finally, crowd support refers to the influence of competing in front of a supportive audience, which has been identified as one of the strongest contributors to home advantage, particularly in events with large attendances.

Additional evidence for the importance of crowd support is provided by Bilalić et al. (2021), who examined sporting events held during the COVID-19 pandemic, when spectators were either absent or severely limited. Their findings showed that the reduction in crowd size substantially weakened the home advantage effect, suggesting that spectator support plays a significant role in improving the performance of home teams. Building on these findings, this study investigates whether hosting the FIFA World Cup provides a measurable competitive advantage for national teams.

A long standing debate in football concerns whether success depends more on offensive strength or defensive solidity. While attacking teams often attract greater attention because of their ability to score goals, defensive organization has traditionally been associated with consistent success. Seippel (2018) examined more than six decades of professional football and found that neither approach is universally superior. Instead, the relative importance of attacking and defensive performance has evolved over time, with modern football showing an increasing emphasis on defensive efficiency. These findings suggest that successful teams must achieve an appropriate balance between scoring goals and preventing their opponents from scoring. Motivated by this debate, this study investigates whether attacking efficiency or defensive strength is more strongly associated with success in FIFA World Cup tournaments.

Research Questions

Question 1: To what extent is GDP per capita associated with a country's FIFA World Cup performance?

Question 2: Does hosting the FIFA World Cup provide a measurable competitive advantage?

Question 3: Which contributes more to World Cup success: attacking efficiency or defensive strength?

Data overview and pre-processing

In order to answer these questions we used 2 datasets, one with the results of all international matches in football history and another one with the gdp per country per year for the period 1960 to 2023. These datasets needed some preprocessing in order for them to be used for this work, we will state in a brief way the preprocessing steps and the justification for them.

Dataset 1: International Football Matches

The original football dataset contains all official international football matches played worldwide. Since the objective of this study is to analyze FIFA World Cup performance, the dataset was filtered to retain only matches corresponding to the FIFA World Cup tournament. This step removed matches from continental championships, qualifiers, friendly matches, and other competitions that were outside the scope of the research.

The match date was then converted from a text format to a date format, allowing the extraction of the tournament year. A new variable containing only the year was created to facilitate aggregation and to serve as the temporal key when merging with the economic dataset.

To simplify the integration process, standardized country names were created for both the home and away teams. Several countries appear under different names across historical records due to political changes or naming conventions ("West Germany" versus "Germany", "Czech Republic" versus "Czechia"). These inconsistencies were corrected by creating auxiliary standardized country name fields, ensuring that equivalent countries were represented by a single identifier throughout the analysis.

Finally, only the variables required for the study were retained, including the participating teams, scores, host country, match location, neutral venue indicator, and tournament year.

Dataset 2: GDP per Capita

The economic dataset originally contained country identifiers, geographic classifications, indicator metadata, GDP per capita values, and annual growth rates. Since the analysis focuses exclusively on GDP per capita, variables unrelated to the research objective including country codes, regional classifications, indicator codes, indicator names, and GDP growth variation were removed to simplify the dataset.

Records with missing GDP per capita values or values equal to zero were eliminated. These observations do not provide meaningful economic information and would prevent valid comparisons between countries.

As with the football dataset, country names were standardized by creating an auxiliary country name field. This ensured that each country could be uniquely matched with the football records despite differences in naming conventions between both data sources.

Data Integration

The two datasets were integrated, following the merge, observations with missing GDP values were inspected individually. Most missing values corresponded to countries that no longer exist (such as Yugoslavia, Czechoslovakia, East Germany, and the Soviet Union), countries with no available historical GDP records, or naming differences that could not be reconciled. Because reliable economic data were unavailable for these cases, they were retained in the match dataset but excluded from analyses requiring GDP per capita.

Feature Engineering

To answer the research questions, several additional variables were derived from the merged dataset.

At the tournament level, team performance metrics were calculated by aggregating all matches played by each country in every FIFA World Cup. These metrics included total matches played, wins, draws, losses, goals scored, goals conceded, goal difference, total points, win percentage, goals scored per match, goals conceded per match, and points per match.

For the analysis of home advantage, a binary variable indicating whether a team was hosting the tournament was created by comparing the participating country's name with the tournament host country. This variable allowed the comparison of team performance during home tournaments against performances in tournaments hosted elsewhere.

Finally, GDP per capita quartiles were computed to classify countries into four economic groups of similar size. This categorization facilitated comparisons of tournament performance among countries with different levels of economic development and supported several of the visualizations presented throughout the study.

Summary of Findings

Before answering the questions it is important to mention that the number of matches has been progressively increasing across time, due to changes in format or the mere increase in participating teams, this explains why in future graphs for some of the early cups information

seems incomplete or not significant, this context will help us understand and justify some of the analysis. We can observe this pattern in figure 1.

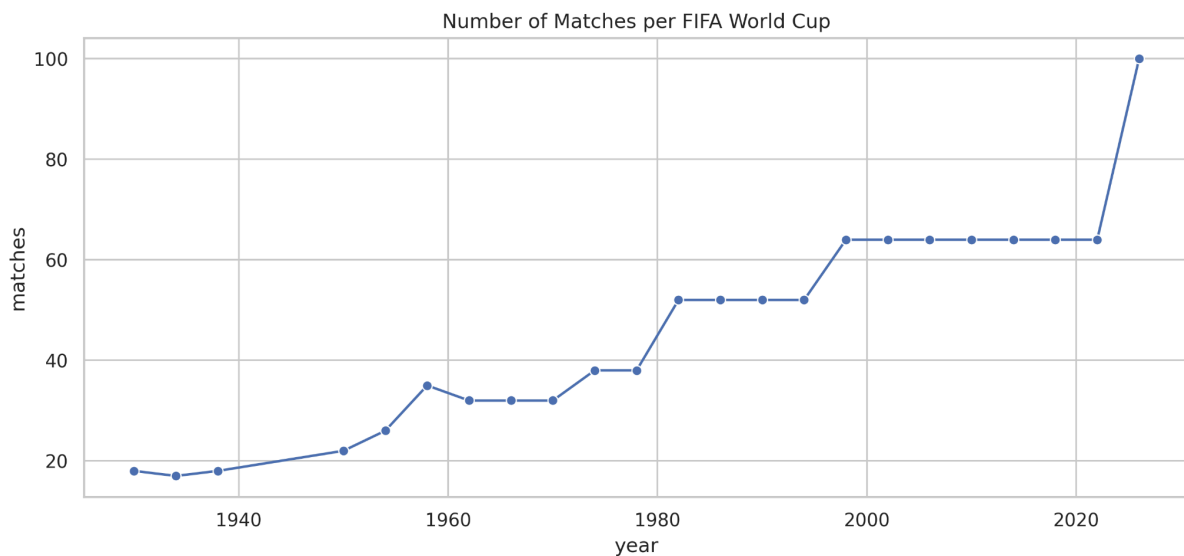


Figure 1: Number of matches in each world cup.

As an interesting fact observed during the EDA, there are eight countries that have won at least one world cup, all of them are in the top ten countries that have played more matches in world cups. The two teams that have never won a world cup but are in this group we encounter Netherlands that have lost 3 finals and interesting enough Mexico has played a lot of games but never reached a semifinal. Figure 2 illustrates this perfectly.

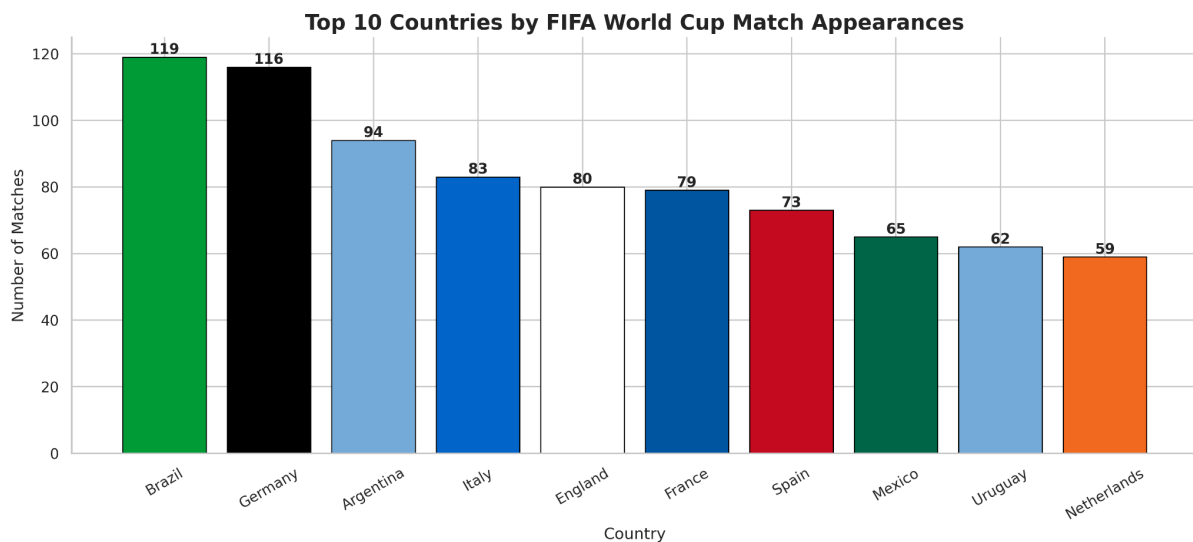


Figure 2: Top 10 countries by number of matches played in world cups.

Q1. To what extent is GDP per capita associated with a country's FIFA World Cup performance?

First we will define performance, performance will be measured in points. There is a convention in football where a win gives 3 points, a lose 0 points and a tie 1 point. Based on this we will measure the performance of each country in points obtained in the tournament. In figure 3 we can observe the amount of points obtained by each country compared to their GDP.

In the first three world cups where we have gdp data there seems to be no correlation in the performance compared to the gdp, but starting in 1974 until 2018 there is a clear trend where countries with higher gdp tend to perform better.

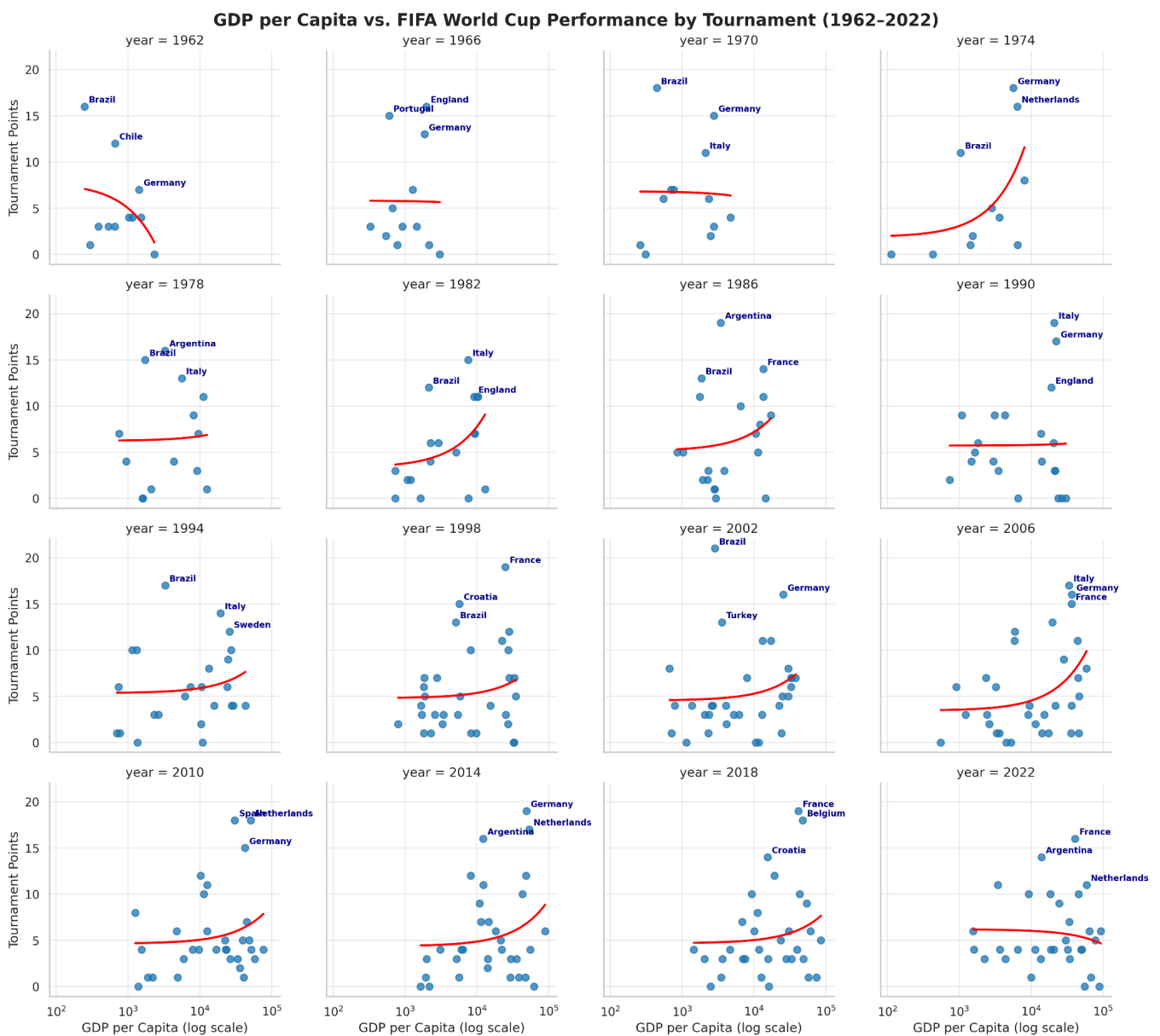


Figure 3: Points versus GDP in all world cups.

It appears that the pattern becomes less evident in the 2022 tournament. To further investigate this relationship, we created an additional visualization grouping countries into GDP per capita quartiles. Figure 4 presents the relationship more clearly, showing that, on average, countries with higher GDP per capita tend to achieve better World Cup performance.

The visualization also highlights several notable outliers. Brazil consistently performed substantially better than would be expected based solely on its GDP per capita, suggesting that factors beyond economic development contribute to its sustained success. Conversely, countries such as New Zealand performed below what their GDP per capita would predict. This may be explained by other factors, such as the popularity of football within the country, the quality of domestic football development programs, or the overall competitiveness of the national team.

Overall, the results indicate that although GDP per capita is not the sole determinant of success, there is a positive association between a country's level of economic development and its average performance in FIFA World Cup tournaments.

Tournament Points by GDP Quartile Across FIFA World Cups (1962-2022)



Figure 4: Points versus GDP quartiles in all world cups.

Q2. Does hosting the FIFA World Cup provide a measurable competitive advantage?

To evaluate the effect of hosting the FIFA World Cup, we maintained the same performance criterion based on points per match. The objective was to compare each host nation's performance during the tournaments it hosted with its average performance in all other World Cup appearances.

Figure 5 shows that 16 of the 19 host nations performed better when playing on home soil than in tournaments hosted elsewhere. Only three countries did not follow this pattern: **Brazil, Qatar, and Spain**. Brazil is a particularly interesting case, as it has won the FIFA World Cup five times, but none of those titles were achieved while hosting the tournament. Nevertheless, in both editions hosted by Brazil (1950 and 2014), the national team still reached at least the semifinals, demonstrating a high level of performance despite not surpassing its historical average.

Qatar represents a different type of outlier. The country qualified automatically as the host in 2022, marking its first World Cup appearance. Consequently, there is only one non host tournament available for comparison (2026), making its average performance outside of hosting less representative. Spain is the remaining exception, although the difference between its home and away performances is relatively small compared with the improvements observed for most other host nations.

Overall, the results indicate that approximately 84% of host countries achieved a higher average number of points per match when hosting the tournament than in their other World Cup participations. These findings provide strong empirical support for the existence of a home advantage in FIFA World Cup competitions.

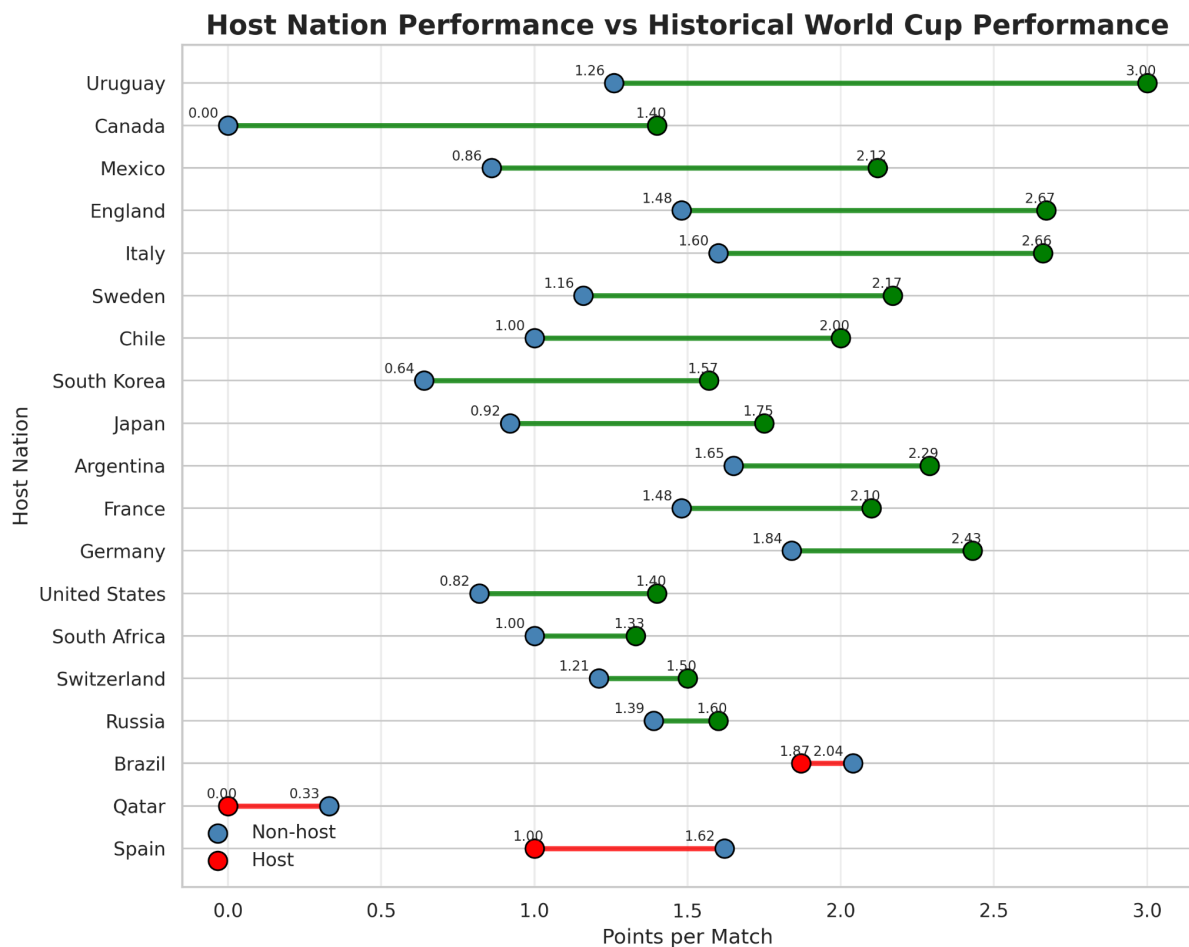


Figure 5: Points per match being host versus non host.

Q3. Which contributes more to World Cup success: attacking efficiency or defensive strength?

When comparing the relative importance of scoring goals versus preventing goals, the correlation analysis shows that both variables are similarly associated with tournament success. As illustrated in Figure 6, goals scored per match exhibits a positive correlation of 0.62 with points obtained, while goals conceded per match presents a negative correlation of -0.60 . The magnitudes of these correlations are remarkably similar, suggesting that neither attacking efficiency nor defensive strength clearly dominates the other as a predictor of success.

These findings are consistent with the literature discussed previously, which argues that successful football performance depends on achieving an appropriate balance between offensive and defensive play, with the relative importance of each varying across different eras and tactical approaches. From a practical perspective, however, scoring goals remains essential, as a team that fails to score cannot earn more than one point from a match.

Finally, Figure 8 illustrates that the team scoring the highest number of goals in a tournament is not necessarily the eventual champion. In fact, in most World Cups, the highest scoring team did not win the title, highlighting that offensive production alone is insufficient to guarantee success. This observation further emphasizes the importance of defensive performance and maintaining a positive goal balance throughout the tournament.

Overall, the results suggest that World Cup success is not driven exclusively by either attacking efficiency or defensive strength. Instead, the evidence indicates that the most successful teams combine effective attacking play with solid defensive organization, achieving a balanced performance on both ends of the game.

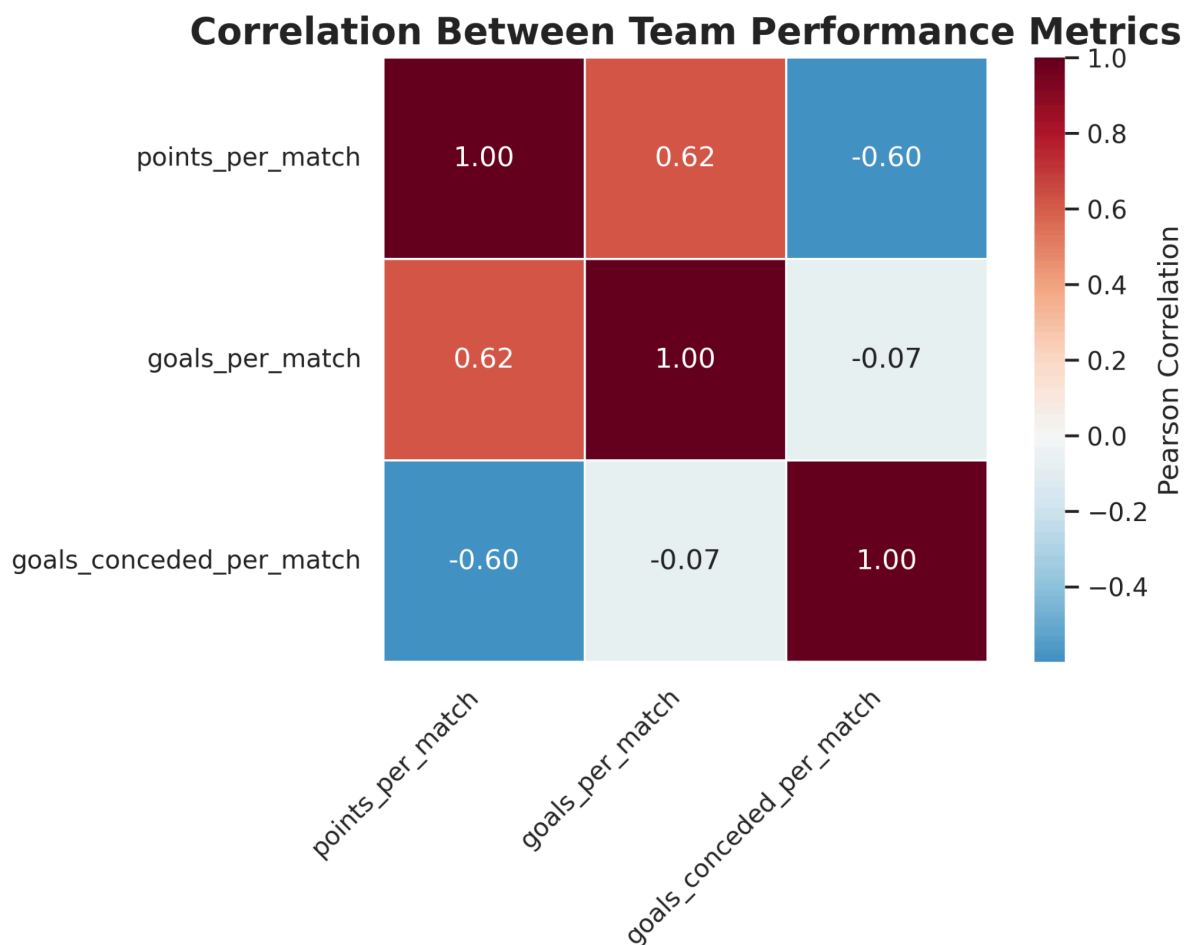


Figure 6: Correlation between points and goals scored and received.

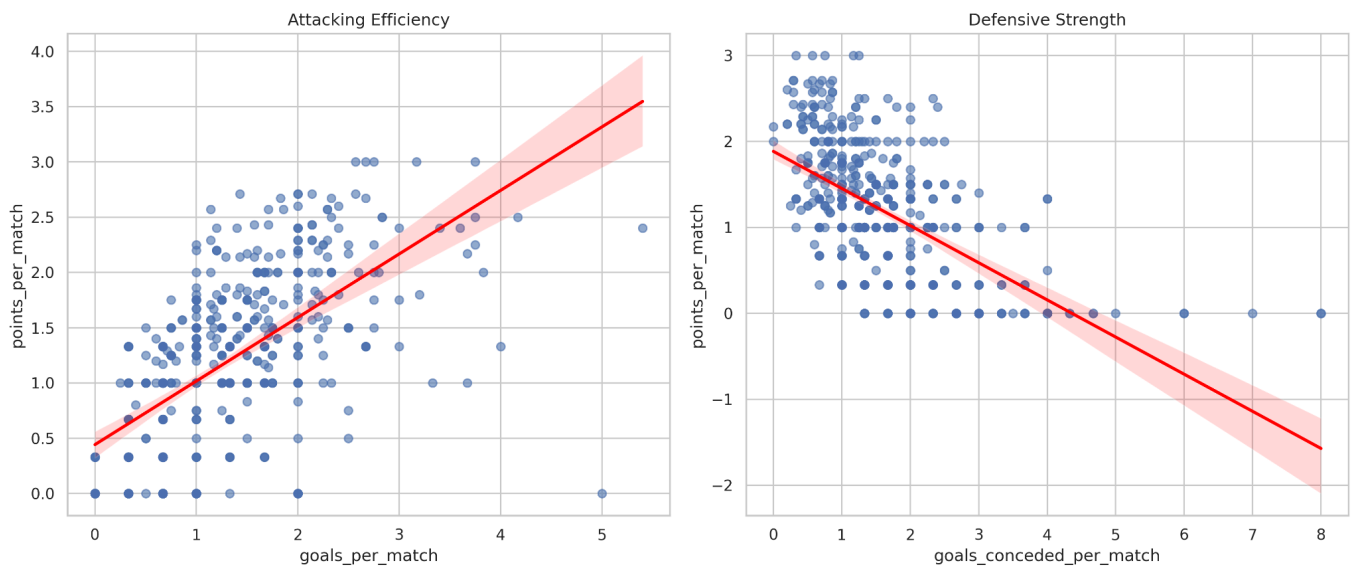


Figure 7: Scatter plots of goals received and scored versus points gained per match

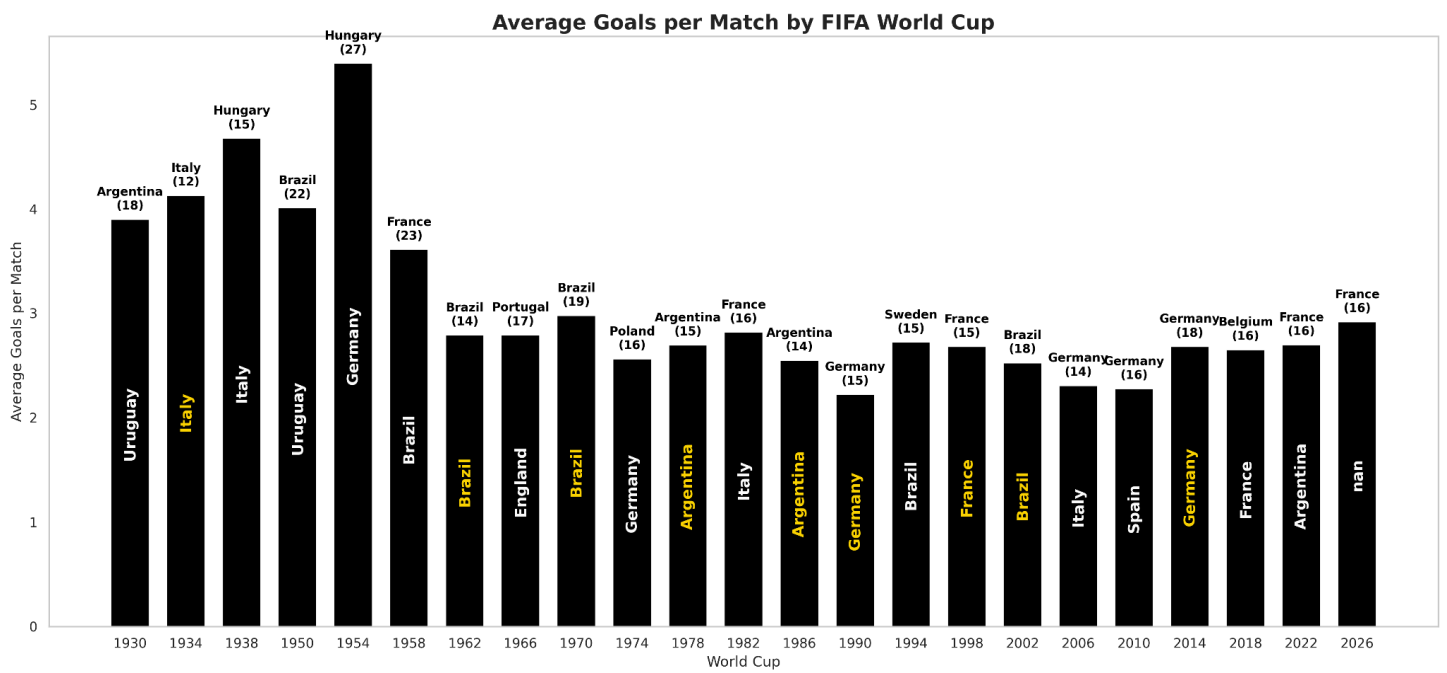


Figure 8: Average goals per game in the world cups with the top team and winner of each cup.

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Appendix

Use of Generative Artificial Intelligence

During the development of this project, ChatGPT (GPT 5.5) was used as a support tool throughout the analysis process. Its use was limited to debugging Python code, obtaining suggestions for data visualization and exploratory analysis, reviewing the clarity and academic style of written sections, and providing feedback on the overall structure of the report.

Bellow are the prompts used:

1. "Review my proposed research questions based on the attached datasets and suggest additional analyses or visualizations that could strengthen the study."
2. "Help me identify why this Python code is producing a KeyError when creating the visualization, and explain how to fix it."
3. "Suggest additional exploratory data analysis (EDA) visualizations that would provide useful insights for my research questions."
4. "How can I automatically export all the Matplotlib graphs generated in my Jupyter Notebook as high resolution image files for inclusion in my report?"

5. "Review this paragraph for grammar, clarity, and academic writing style without changing the meaning of my analysis."
6. "Based on the visualizations and results obtained so far, suggest other figures, comparisons, or metrics that could add value to the paper."
7. "Recommend relevant academic sources discussing GDP and sports performance, home advantage in football, and the relationship between attacking and defensive efficiency, so that I can review and cite the appropriate literature."